Decentralized IoT Edge-Swarm Task Allocation & Self-Healing Mesh Simulator

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Abstract

An observation is useful only when a valid product reaches its destination before its deadline. In a satellite swarm, this requires a feasible chain of acquisition, processing, storage and communication despite changing resources and faults. We formulate this chain and implement a reduced simulator combining Earth-centred J_2 propagation, annual Sun geometry, eclipse-dependent charging, CPU and thermal limits, mounted-antenna access and acknowledged forwarding. A temporary onboard coordinator buffers requests and shares task manifests. The alternating direction method of multipliers (ADMM) ranks hosts through a convex admission relaxation; independent checks construct integer assignments. A bounded particle swarm optimization (PSO) search repairs selected unfinished tasks and chooses among contact-route alternatives. Temperature residuals, telemetry age and hard resource limits trigger capability-specific isolation. The archived 6000-s run records 40 delivered tasks, 542 expired tasks and eight active tasks, with 34 reroutes. These results expose delivery-contact limits rather than establish an advantage over other algorithms. The present program emulates the agents and ownership authority in one process; distributed replication, calibrated hardware models and comparative trials remain future work.

Keywords: CubeSat swarm; task admission; ADMM; fault recovery; particle swarm optimization; contact routing; edge computing.

1 Introduction

Satellite and industrial sensor networks must often react before a ground station or cloud controller becomes reachable. New requests arrive while processors are busy, batteries are charging or discharging, and communication opportunities change. Assigning work using battery level alone can select a node that cannot point, store the data or deliver the result. A useful allocator must therefore evaluate the complete task chain.

Distributed optimization separates local resource constraints from decisions shared across agents [3, 5]. Local schedule repair limits the amount of work reconsidered after an event. Shen et al. apply this idea to dynamic UAV allocation [1]; the present implementation adapts the principle to orbital opportunities and retained data. A satellite cannot repair a missed observation by moving directly towards the target. Its choices concern available observation windows, processing capacity and future contacts. Delay-tolerant networking supplies the corresponding store-and-forward view [8, 7].

This paper follows the information from request arrival to ground acknowledgement. The models explain which actions are possible; ADMM supplies fractional preferences; integer admission and PSO construct bounded repairs; ownership checks authorize assignments; and execution updates the resource and data ledgers. Fault handling changes the capabilities available to these stages rather than treating every component fault as total spacecraft loss.

The contribution is an integrated, inspectable simulator and its mathematical formulation. The implementation is a single-process emulation of a workflow intended for onboard coordination. We distinguish its implemented equations from a more complete reference model wherever that distinction affects the claims. No global mission optimum, flight qualification or distributed agreement guarantee is claimed.

2 Models

2.1 State, tasks and completion

Let $\mathcal{V} = \{1, \dots, N\}$ be the satellites, g the ground sink, k a simulation sample, and $\mathcal{J}$ the current tasks. The reduced state is

$$s_{i,k} = \left(\boldsymbol{r}_i, \boldsymbol{v}_i, R_i, E_i, T_i, u_i, B_i^{\text{used}}, B_i^{\text{res}}, \boldsymbol{h}_i, t_i^{\text{tel}}\right)_k. \tag{2.1}$$

Here R_i contains body-axis directions, E_i is battery energy in Wh, T_i is temperature in °C, u_i is CPU usage in percentage points, and $\boldsymbol{h}_i$ describes component capabilities. Position and velocity use km and km/s. Storage uses decimal MB, so $1\,\text{MB} = 10^6$ bytes.

A task is specified by

$$\tau_j = \left(\text{id}_j, \mathcal{G}_j, [a_j, b_j], d_j, p_j, \delta_j, C_j, D_j^{\text{raw}}, D_j^{\text{out}}, q_j^{\min}, u_j\right), \tag{2.2}$$

where $\mathcal{G}_j$ is the target, $[a_j, b_j]$ the observation window, d_j the delivery deadline, p_j the priority, δ_j the total required capture duration, C_j the processing cycles, and u_j the reserved CPU percentage. Repeated surveillance generates a new observation-instance ID. Changing the priority of an existing event preserves its task IDs.

An assignment has owner o_j; an acquired data product has holder h_j. These variables have different meanings. Delivery success is

$$z_j^{\text{del}} = \mathbf{1}\left\{t_j^{\text{ack}} \le d_j, \quad q_j \ge q_j^{\min}, \quad \text{complete product acknowledged by } g\right\}. \tag{2.3}$$

Capture and processing alone earn no delivered-product count. The normal states are pending, assigned, captured, processed and delivered; deferral, expiry and failure record alternatives explicitly.

2.2 Orbital propagation and perturbation

The model needs satellite position to predict eclipse, antenna access and ground contacts. In the display coordinates, the y axis is the Earth polar axis and x, z span the equatorial plane. With $r = \|\boldsymbol{r}\|$,

$$\dot{\boldsymbol{r}}_i = \boldsymbol{v}_i, \tag{2.4}$$

$$\dot{\boldsymbol{v}}_i = -\frac{\mu}{r_i^3}\boldsymbol{r}_i + \boldsymbol{a}_{J_2}(\boldsymbol{r}_i), \tag{2.5}$$

$$\boldsymbol{a}_{J_2} = \frac{3J_2\mu R_J^2}{2r^5}\begin{bmatrix} x(5y^2/r^2 - 1) \\ y(5y^2/r^2 - 3) \\ z(5y^2/r^2 - 1) \end{bmatrix}. \tag{2.6}$$

The constants are $\mu = 398600.4418\,\text{km}^3/\text{s}^2$, $J_2 = 1.08262668 \times 10^{-3}$ and $R_J = 6378.137$ km. The spherical occulting Earth has radius $R_E = 6371$ km. These radii serve different purposes. The initial altitude is 550 km and inclination $\iota = 51.6°$. Adjacent satellites start with chord spacing $s = 50$ km:

$$r_0 = R_E + 550, \qquad \Delta\nu = 2\arcsin\frac{s}{2r_0}, \qquad \nu_i = \nu_c + \left(i - \frac{N-1}{2}\right)\Delta\nu. \tag{2.7}$$

For zero-based i, the initial vectors are

$$\boldsymbol{r}_i(0) = R_y(\Omega) r_0 \begin{bmatrix} \cos\nu_i \\ \sin\nu_i \sin\iota \\ \sin\nu_i \cos\iota \end{bmatrix}, \tag{2.8}$$

$$\boldsymbol{v}_i(0) = R_y(\Omega)\sqrt{\frac{\mu}{r_0}}\begin{bmatrix} -\sin\nu_i \\ \cos\nu_i \sin\iota \\ \cos\nu_i \cos\iota \end{bmatrix}, \qquad R_y(\Omega) = \begin{bmatrix} \cos\Omega & 0 & -\sin\Omega \\ 0 & 1 & 0 \\ \sin\Omega & 0 & \cos\Omega \end{bmatrix}. \tag{2.9}$$

The common phase and plane place the formation near the ground station at the chosen initial epoch. The two named swarms are groups in this common formation, not separate orbital planes.

Fourth-order Runge–Kutta integrates equation (2.5). The main simulation advances in steps $\Delta t \le 1$ s, including at accelerated display speeds. The model includes no thrust, atmospheric drag, Moon or Sun gravity. J_2 changes the propagated orbit; it does not maintain the initial spacing.

2.3 Earth rotation, revolution and solar exposure

Earth rotation changes the ground station's inertial position. For latitude ϕ, longitude λ and rotation rate ω_E,

$$\boldsymbol{r}_g(t) = R_E \begin{bmatrix} \cos\phi\cos(\lambda+\omega_E t) \\ \sin\phi \\ \cos\phi\sin(\lambda+\omega_E t) \end{bmatrix}, \qquad \omega_E = \frac{2\pi}{86164.0905\,\mathrm{s}}. \tag{2.10}$$

The station is at 18.52° N, 73.86° E. A circular annual model gives the unit vector towards the distant Sun:

$$\hat{\boldsymbol{s}}(t) = \begin{bmatrix} \cos\lambda_\odot \\ \sin\lambda_\odot\sin\epsilon \\ \sin\lambda_\odot\cos\epsilon \end{bmatrix}, \quad \lambda_\odot = \frac{2\pi t}{365.256363004\times 86400}, \quad \epsilon = 23.43928^\circ. \tag{2.11}$$

The epoch is an illustrative equinox phase. Annual revolution changes the Sun direction; satellite revolution changes the satellite position; Earth spin changes ground access. Solar rays are parallel with constant irradiance $S_0 = 1361\,\mathrm{W/m^2}$. Annual inverse-square modulation is omitted.

A cylindrical shadow declares eclipse when the satellite lies behind Earth and within its shadow radius:

$$\chi_i^{\mathrm{ecl}} = \mathbf{1}\left\{\boldsymbol{r}_i^\top\hat{\boldsymbol{s}} < 0, \quad r_i^2 - (\boldsymbol{r}_i^\top\hat{\boldsymbol{s}})^2 < R_E^2\right\}, \qquad \chi_i^\odot = 1 - \chi_i^{\mathrm{ecl}}. \tag{2.12}$$

For body-axis unit vectors $\boldsymbol{b}_{i,1}, \boldsymbol{b}_{i,2}, \boldsymbol{b}_{i,3}$, the implemented two-sided solar-wing model is

$$P_i^\odot = \chi_i^\odot S_0(0.14)(0.28)\max\left(0.08, |\boldsymbol{b}_{i,2}^\top\hat{\boldsymbol{s}}|\right). \tag{2.13}$$

Area is in m^2 and efficiency is dimensionless. The incidence floor is a demonstration approximation. It is not an exact flat-panel response. The direct absorbed heat model is

$$Q_i^\odot = \chi_i^\odot S_0(0.018)(0.6)\left(0.5|\boldsymbol{b}_{i,1}^\top\hat{\boldsymbol{s}}| + 0.3|\boldsymbol{b}_{i,2}^\top\hat{\boldsymbol{s}}| + 0.2|\boldsymbol{b}_{i,3}^\top\hat{\boldsymbol{s}}|\right). \tag{2.14}$$

Thus eclipse removes both charging and direct solar heating. The same body orientation affects the antennas, so power and communication are coupled.

2.4 CPU usage and power

Let u_i^{ctl}, u_i^{stress} and u_i^{alloc} denote control, injected stress and task reservations. The available CPU is

$$u_i^{\mathrm{free}} = \left[100 - u_i^{\mathrm{ctl}} - u_i^{\mathrm{stress}} - u_i^{\mathrm{alloc}}\right]_+, \qquad \sum_{j\in\mathcal{J}_i^{\mathrm{res}}} u_j \le 100 - u_i^{\mathrm{ctl}} - u_i^{\mathrm{stress}}. \tag{2.15}$$

During processing, remaining work $C_j - c_j$ requires approximately

$$t_{ij}^{\mathrm{proc}} = \frac{C_j - c_j}{f_i u_j/100}, \qquad \dot{c}_j = f_i\frac{u_j}{100}\min\left(1, \frac{\left[100 - u_i^{\mathrm{ctl}} - u_i^{\mathrm{stress}}\right]_+}{\max(1, u_i^{\mathrm{alloc}})}\right). \tag{2.16}$$

Here $\mathcal{J}_i^{\mathrm{res}}$ contains tasks with active CPU reservations. The throttle factor protects available processor capacity when stress rises. The simulation samples f_i between 1.6 and 2.4 million cycles/s. Processing remains on the observer; migration of raw images is not implemented.

The reduced electrical load is

$$P_i^{\mathrm{base}} = 8 + 0.45u_i^{\mathrm{actual}} + P_i^{\mathrm{extra}} \quad \mathrm{W}, \qquad u_i^{\mathrm{actual}} = \min(100, u_i^{\mathrm{ctl}} + u_i^{\mathrm{stress}} + u_i^{\mathrm{alloc}}). \tag{2.17}$$

Capture adds 12 W over its active duration. Transmitting one MB costs 0.002 Wh at the sender. These coefficients define the demonstration workload, not a measured flight computer or radio.

2.5 Thermal management and battery levels

A physical thermal reference is

$$C_i^{\mathrm{th}}\dot{T}_{i,K} = Q_i^{\mathrm{diss}} + Q_i^\odot + Q_i^{\mathrm{alb}} + Q_i^{\mathrm{IR}} + Q_i^{\mathrm{cond}} - \varepsilon_i\sigma A_i(T_{i,K}^4 - T_{\mathrm{space,K}}^4). \tag{2.18}$$

Radiative temperatures are in kelvin. There is no ambient convective cooling in orbit. The implementation instead uses a local linear approximation to heat rejection, consistent with a reduced demonstration model [11, 12]:

$$C^{\mathrm{th}}\dot{T}_i = Q_i - K_i(T_i - T_{\mathrm{ref}}), \qquad Q_i = 4 + 0.50u_i + 5\chi_i^{\mathrm{radio}} + Q_i^{\mathrm{extra}} + Q_i^{\odot}, \tag{2.19}$$

$$T_i(t+\Delta t) = T_i^{\star} + (T_i(t) - T_i^{\star})e^{-K_i\Delta t/C^{\mathrm{th}}}, \qquad T_i^{\star} = T_{\mathrm{ref}} + Q_i/K_i. \tag{2.20}$$

Here $C^{\mathrm{th}} = 20\,\mathrm{J/K}$, $K_i = 0.9\kappa_i\,\mathrm{W/K}$, $T_{\mathrm{ref}} = 20°\mathrm{C}$ and κ_i is the cooling multiplier. The small thermal capacity deliberately makes faults visible quickly. Actual temperature updates use $\chi_i^{\mathrm{radio}} = 0$; the admission forecast uses one as a conservative allowance. Individual radio and payload heat pulses are therefore not fully resolved.

The battery balance is

$$E_{i,k+1} = \mathrm{clip}_{[0,E_i^{\max}]}\left[E_{i,k} + \frac{(0.9P_i^{\odot} - P_i^{\mathrm{base}})\Delta t}{3600} - \frac{12\Delta t_i^{\mathrm{cap}}}{3600} - 0.002D_{i,k}^{\mathrm{tx}}\right]. \tag{2.21}$$

The factor 3600 converts W s to Wh. Battery capacity is $E_i^{\max} = 100$ Wh and the protection reserve is $E^{\mathrm{res}} = 12$ Wh. Clipping bounds the numerical state; it is not evidence that a schedule met its reserve constraint.

For admission, define $P_i^0 = 8 + 0.45(u_i^{\mathrm{ctl}} + u_i^{\mathrm{stress}} + u_i^{\mathrm{alloc}})$. The implemented task-energy proxy is

$$\widehat{e}_{ij} = \frac{P_i^0(\delta_j + t_{ij}^{\mathrm{proc}} + 20) + 12\delta_j + 0.45u_j t_{ij}^{\mathrm{proc}}}{3600} + 0.006D_j^{\mathrm{out}}. \tag{2.22}$$

It credits no future solar power. Once a route predicts arrival A_{ij}, the admission guard requires

$$E_i - \widehat{e}_{ij} - \frac{P_i^0\,[A_{ij} - t]_+}{3600} \geq E^{\mathrm{res}} + 0.2\ \mathrm{Wh}. \tag{2.23}$$

This proxy includes overlapping baseline allowances. It is a screening rule, not an exact future energy trajectory. A separate constant-load thermal forecast must satisfy $\widehat{T}_i \leq 76 - 2 = 74°\mathrm{C}$. Hardware-calibrated generation, conversion losses and mission-specific reserve prediction are needed for operational use [10, 12].

2.6 Storage and observation eligibility

Reservations prevent an apparently empty buffer from being promised to several tasks. The available memory is

$$B_i^{\mathrm{free}} = B_i^{\max} - B_i^{\mathrm{used}} - B_i^{\mathrm{res}}, \qquad B_i^{\mathrm{used}} + B_i^{\mathrm{res}} \leq B_i^{\max}. \tag{2.24}$$

An assigned task reserves $D_j^{\mathrm{raw}} + D_j^{\mathrm{out}}$. After capture, raw data is used memory and output remains reserved. After processing, the output replaces the raw buffer. A relay reserves the full output before receiving it. The old holder releases its copy only after the next holder acknowledges the complete product. The default capacity is 512 MB.

Let $\mathcal{W}_{ij}$ contain access intervals for host i. A valid capture start satisfies

$$t_{ij}^{\mathrm{cap}} = \min\left\{s \geq t : [s, s+\delta_j] \subseteq [a_j, b_j] \cap [w^-, w^+], \quad [w^-, w^+] \in \mathcal{W}_{ij}\right\}. \tag{2.25}$$

The present windows are synthetic: 85-s intervals repeat every 110 s with node phase $7i \bmod 110$ s, clipped to the requested window. Target coordinates are metadata; the code does not yet derive camera access from them. The reference geometric gate would instead be

$$\chi_{ij}^{\mathrm{obs}} = \chi^{\mathrm{LOS}}\chi^{\mathrm{FOV}}\chi^{\mathrm{point}}\chi^{\mathrm{illum}}\chi^{\mathrm{quality}}\chi^{\mathrm{health}}. \tag{2.26}$$

In the implemented gate, a host needs an explicit window, fresh telemetry, CPU and storage capacity, acceptable energy and temperature, attitude error at most 5°, and a predicted delivery path. Acquired quality is fixed at $q_j = 0.96$; requests exceeding that value are rejected. This separates testing of the coordination logic from a future optical-quality model.

2.7 Antennas and link graphs

For endpoints i, l, let $d_{il} = \|\boldsymbol{r}_l - \boldsymbol{r}_i\|$ and $\hat{\boldsymbol{d}}_{il} = (\boldsymbol{r}_l - \boldsymbol{r}_i)/d_{il}$. Earth line of sight is checked at the closest point on the connecting segment:

$$\zeta^* = \mathrm{clip}_{[0,1]}\left(-\frac{\boldsymbol{r}_i^\top(\boldsymbol{r}_l - \boldsymbol{r}_i)}{d_{il}^2}\right), \qquad \chi_{il}^{\mathrm{LOS}} = \mathbf{1}\{\|\boldsymbol{r}_i + \zeta^*(\boldsymbol{r}_l - \boldsymbol{r}_i)\| \geq R_E\}. \tag{2.27}$$

A small tolerance is used numerically at the Earth surface. Each spacecraft is a $0.10 \times 0.10 \times 0.30$ m box with fore and aft antenna mounts; ground links may also use a nadir mount. The line direction in body coordinates is $\hat{\boldsymbol{d}}_{il}^B = R_i^\top \hat{\boldsymbol{d}}_{il}$. For mount normal $\boldsymbol{n}_a$,

$$\theta_a = \arccos(\boldsymbol{n}_a^\top \hat{\boldsymbol{d}}_{il}^B), \qquad g_a = \mathbf{1}\{\theta_a \leq 72^\circ,\ \text{ray clears body}\}\max(0, \cos\theta_a)^4. \tag{2.28}$$

Ray–box intersection tests $\boldsymbol{o}_a + \ell\hat{\boldsymbol{d}}_{il}^B$ for $\ell > 0$ against the three box slabs. For nonzero direction components,

$$\ell_{\mathrm{enter}} = \max_m \min\left(\frac{-a_m - o_{a,m}}{d_m^B}, \frac{a_m - o_{a,m}}{d_m^B}\right), \quad \ell_{\mathrm{exit}} = \min_m \max\left(\frac{-a_m - o_{a,m}}{d_m^B}, \frac{a_m - o_{a,m}}{d_m^B}\right). \tag{2.29}$$

An intersection blocks the ray when $\ell_{\mathrm{exit}} \geq \max(\ell_{\mathrm{enter}}, 10^{-6})$; zero direction components use the parallel-slab test. The best usable mount determines each endpoint's gain.

A link exists only when Earth visibility, both antenna gates, endpoint radio capability and outage policy permit it. For ground access, elevation must also exceed 5°:

$$\frac{(\boldsymbol{r}_i - \boldsymbol{r}_g)^\top \boldsymbol{r}_g}{\|\boldsymbol{r}_i - \boldsymbol{r}_g\| R_E} \geq \sin 5^\circ. \tag{2.30}$$

The coarse model is geometric; it is not an electromagnetic link budget. Propagation delay and a Doppler diagnostic are

$$L_{il} = \frac{d_{il}}{c} + 0.003\ \mathrm{s}, \qquad f_{D,il} = -\frac{(\boldsymbol{v}_l - \boldsymbol{v}_i)^\top \hat{\boldsymbol{d}}_{il}}{c} f_c, \quad f_c = 2.2\,\mathrm{GHz}, \quad c = 299792.458\,\mathrm{km/s}. \tag{2.31}$$

For a seeded symmetric calibration $\xi_{il} \in [0, 1)$ and $\bar{g}_{il} = \max(0.015, \sqrt{g_{i\to l} g_{l\to i}})$, the link cost is

$$\begin{aligned} w_{il} = {} & 0.4 + 1.2\xi_{il} + 0.4\frac{d_{il}}{1000\,\mathrm{km}} + 1.3(\bar{g}_{il}^{-1} - 1) \\ & + 0.3(1 - E_i/100) + 0.3(1 - E_l/100) + 0.2\frac{|f_{D,il}|}{10^5\,\mathrm{Hz}} + 25(n_i^{\mathrm{block}} + n_l^{\mathrm{block}}), \end{aligned} \tag{2.32}$$

$$R_{il} = (18 + 10\xi_{il})\max(0.12, \bar{g}_{il}) \quad \mathrm{Mbit/s}. \tag{2.33}$$

The ground endpoint contributes no battery penalty. All terms in w_{il} are dimensionless demonstration scores; lower is preferred. An unavailable edge remains unavailable regardless of its score. The rate is a surrogate, not a Shannon-capacity or measured modulation model.

2.8 Coupled mission objective

Let $y_{ij} \in \{0, 1\}$ admit task j to host i and let Π collect timing, processing and routing choices. A reference objective is

$$\max_{\Pi, y} \sum_j p_j q_j z_j^{\mathrm{del}} - \lambda_E \sum_i \frac{e_i(\Pi)}{E_i^{\max}} - \lambda_L \sum_j \frac{[A_j - a_j]_+}{\max(1, d_j - a_j)} - \lambda_\Delta\,\mathrm{Changes}(\Pi, \Pi^{\mathrm{old}}), \tag{2.34}$$

subject to $\sum_i y_{ij} \leq 1$, $z_j^{\mathrm{del}} \leq \sum_i y_{ij}$ and the resource, access, precedence and contact constraints above. Impossible work is deferred or expired; its physical constraints are not relaxed. The simulator solves a reduced admission problem followed by online execution, not this entire joint objective at every tick.

3 Algorithms

3.1 ADMM ranking and integer admission

Freeze a batch of at most 16 queued tasks and the current resource estimates. For each host, let $x_i \in \mathbb{R}^M$ contain relaxed admissions and let $a_{ij} \in \{0,1\}$ be one only when the complete-chain gate passes and that host has received the manifest. The local convex set is

$$\mathcal{F}_i = \{x_i : 0 \le x_{ij} \le a_{ij},\ A_i x_i \le b_i\}, \tag{3.1}$$

$$(A_i)_{:,j} = \begin{bmatrix} u_j/100 \\ (D_j^{\mathrm{raw}} + D_j^{\mathrm{out}})/B_i^{\max} \\ \widehat{e}_{ij}/E_i^{\max} \end{bmatrix}, \qquad b_i = \begin{bmatrix} u_i^{\mathrm{free}}/100 \\ B_i^{\mathrm{free}}/B_i^{\max} \\ [E_i - E^{\mathrm{res}}]_+/E_i^{\max} \end{bmatrix}. \tag{3.2}$$

The integrated implementation has these three additive rows. Thermal and deadline feasibility enter through a_{ij} and the later independent check; they are not exact joint time-indexed constraints in A_i.

The implemented host cost is

$$\begin{aligned} c_{ij} = {} & 0.5(1 - E_i/100) + 0.35(u_i^{\mathrm{alloc}} + u_i^{\mathrm{stress}})/100 + 0.2\frac{[T_i - 40]_+}{30} \\ & + 0.04W_{ij} + 0.2\widehat{e}_{ij} + 0.15\frac{A_{ij} - t}{\max(1, d_j - t)} - (1.2 + 0.5p_j), \end{aligned} \tag{3.3}$$

where W_{ij} is the predicted route's summed link cost. The coefficient of $\widehat{e}_{ij}$ carries reciprocal-Wh scaling. Priority makes useful tasks attractive; energy, heat, congestion and delay make expensive hosts less attractive. These tuning weights are empirical choices.

Introduce a copy Z with per-task coupling set $\mathcal{C} = \{Z : Z \ge 0,\ \sum_i Z_{ij} \le 1\}$. The two-block relaxation is

$$\min_{X,Z} \sum_i \left(c_i^\top x_i + \frac{\gamma}{2}\|x_i\|_2^2 + I_{\mathcal{F}_i}(x_i) \right) + I_{\mathcal{C}}(Z), \qquad X = Z, \quad \gamma = 0.3, \tag{3.4}$$

where I is zero inside a set and $+\infty$ outside. The scaled ADMM updates with penalty $\rho = 2$ are [5]

$$x_i^{\ell+1} = \mathrm{Proj}_{\mathcal{F}_i}\left(\frac{\rho(z_i^\ell - u_i^\ell) - c_i}{\rho + \gamma} \right), \tag{3.5}$$

$$Z_{:,j}^{\ell+1} = \mathrm{Proj}_{\{z \ge 0,\, \mathbf{1}^\top z \le 1\}}(X_{:,j}^{\ell+1} + U_{:,j}^{\ell}), \tag{3.6}$$

$$U^{\ell+1} = U^\ell + X^{\ell+1} - Z^{\ell+1}. \tag{3.7}$$

Here U is a scaled multiplier, unrelated to CPU percentage u_i. The local projection uses Dykstra's method over the box and resource halfspaces. The task projection has form $z_i = \max(v_i - \theta, 0)$, where $\theta = 0$ if the nonnegative entries already sum to at most one; otherwise θ makes their sum one.

Residuals measure disagreement and change:

$$r^{\ell+1} = \|X^{\ell+1} - Z^{\ell+1}\|_F, \qquad s^{\ell+1} = \rho\|Z^{\ell+1} - Z^\ell\|_F. \tag{3.8}$$

The solver stops when both are at most 0.01 or after 20 iterations. The dashboard reports whether the tolerances were reached. A finite iterate is a ranking proposal, including when the solver has not converged.

Integer admission processes repair tasks first, then decreasing priority, earlier deadline and task ID. Hosts are ranked lexicographically by

$$(-Z_{ij},\ c_{ij},\ \mathrm{id}_i)\,. \tag{3.9}$$

For each candidate, the program repeats the complete-chain check using the updated ledgers, checks ownership, and reserves resources before considering the next task. Independent rounding is avoided because it can overload a host or duplicate an exclusive assignment. ADMM convergence applies to the fixed convex relaxation under its assumptions, not to the changing integer mission [5, 3].

3.2 Mesh healing using bounded particle swarm optimization

A fault can release unfinished acquisitions and invalidate product routes. The graph first removes unavailable capabilities and contacts. PSO then searches a small repair set; it cannot create a physically

missing link. Shortest path with nonnegative weights is polynomial-time solvable. The combinatorial difficulty lies in jointly assigning multiple tasks under resource, timing and routing constraints.

At most eight repair tasks form a particle. Coordinate $\alpha_{pj} \in [0, 1)$ selects a starting host in task j's feasible ranked list $\mathcal{H}_j$:

$$m_{pj} = \lfloor \alpha_{pj} |\mathcal{H}_j| \rfloor. \tag{3.10}$$

The decoder tries hosts from m_{pj} cyclically and debits copied CPU, storage and energy ledgers. Its score is

$$F(\alpha_p) = \sum_{j \text{ placed}} (10 p_j + Z_{o_j j}). \tag{3.11}$$

Only tentative placements are scored; each actual commit is revalidated by the full admission routine. This final check covers constraints simplified inside the fast decoder.

With independent uniform vectors $\boldsymbol{r}_1, \boldsymbol{r}_2$, the random-key update is

$$\boldsymbol{v}_p^{\ell+1} = 0.55 \boldsymbol{v}_p^{\ell} + 1.35 \boldsymbol{r}_1 \odot (\boldsymbol{\alpha}_p^{\text{best}} - \boldsymbol{\alpha}_p^{\ell}) + 1.35 \boldsymbol{r}_2 \odot (\boldsymbol{\alpha}^{\text{best}} - \boldsymbol{\alpha}_p^{\ell}), \tag{3.12}$$

$$\boldsymbol{\alpha}_p^{\ell+1} = \text{clip}_{[0, 0.999999]} \left(\boldsymbol{\alpha}_p^{\ell} + \text{clip}_{[-0.35, 0.35]} (\boldsymbol{v}_p^{\ell+1}) \right). \tag{3.13}$$

The default budget is six particles and 12 iterations, giving

$$N_{\text{eval}} = P(K_P + 1) = 6(12 + 1) = 78. \tag{3.14}$$

One initial particle is all zeros, preserving the ADMM-ranked greedy incumbent on the same frozen decoder problem. The search retains the best score found. This implements a bounded local adaptation of the PSO family [6]; its fixed coefficients and random-key encoding differ from Shen et al.'s adaptive UAV schedule search [1].

The locked part consists of acquired data, processing progress and unaffected commitments. The adjustable part consists of eligible unfinished tasks:

$$\Pi = \Pi^{\text{locked}} \oplus \Pi^{\text{free}}. \tag{3.15}$$

Safety checks can cancel a previously committed action. Prefix locking limits unnecessary replanning; it does not override protection.

3.3 Contact graph rerouting

A contact is $e = (i, l, t_e^-, t_e^+, R_e, L_e, w_e)$: endpoints, opening and closing times, rate, latency and cost. Inter-satellite contacts use currently available links over the short planning horizon. Ground windows are predicted from station rotation and a circular short-horizon orbital approximation, sampled every five seconds. Actual execution uses the current J_2 state and antenna gates.

For product size D MB available at i at time A_i, the routing estimate uses

$$t_e^{\text{start}} = \max(A_i, t_e^-), \tag{3.16}$$

$$R_e^{\text{eff}} = \min(24, R_e)\,(0.48)\max(0.25, 1 - \ell_e), \tag{3.17}$$

$$A_l = t_e^{\text{start}} + \frac{8D}{R_e^{\text{eff}}} + 2L_e + 1 \text{ s}, \tag{3.18}$$

$$A_l \le \min(t_e^+, d_j). \tag{3.19}$$

Here ℓ_e is the modeled loss probability; the factors provide retry and sharing allowances. The path must be causal, avoid repeated nodes, and have enough available receiver memory. Waiting for a future ground contact is allowed. Earliest-arrival label search supplies the first route. Up to two alternatives are generated by banning edges of that route, giving at most three distinct candidates.

For repair, PSO selects from those candidates using

$$C(p) = A_g(p) - t + 2W(p), \qquad W(p) = \sum_{e \in p} w_e, \tag{3.20}$$

where the coefficient two converts the dimensionless link score into a time-weighted preference. For only three candidates, direct evaluation of all candidates is simpler and exact over that list. PSO is retained

as an experimental option; it provides no guarantee of finding even the best list member within a finite stochastic search.

This is simplified contact-aware store-and-forward routing, inspired by CGR [7]. It does not implement the complete Bundle Protocol or a standards-compliant CGR/SABR stack. Glowworm-based mesh search [2] motivates alternative heuristic routing experiments, but GSO is not executed in this version. Neither PSO nor GSO can bridge a physical partition without a future contact.

4 Architecture

4.1 Global workflow and architecture

The Python backend holds authoritative simulation state. The browser displays positions, capabilities, tasks, ledgers and graph history; UI actions submit events to the backend. The agent states and control-message buffers are separate objects, although they execute within one process. Figure 1 summarizes the closed loop.

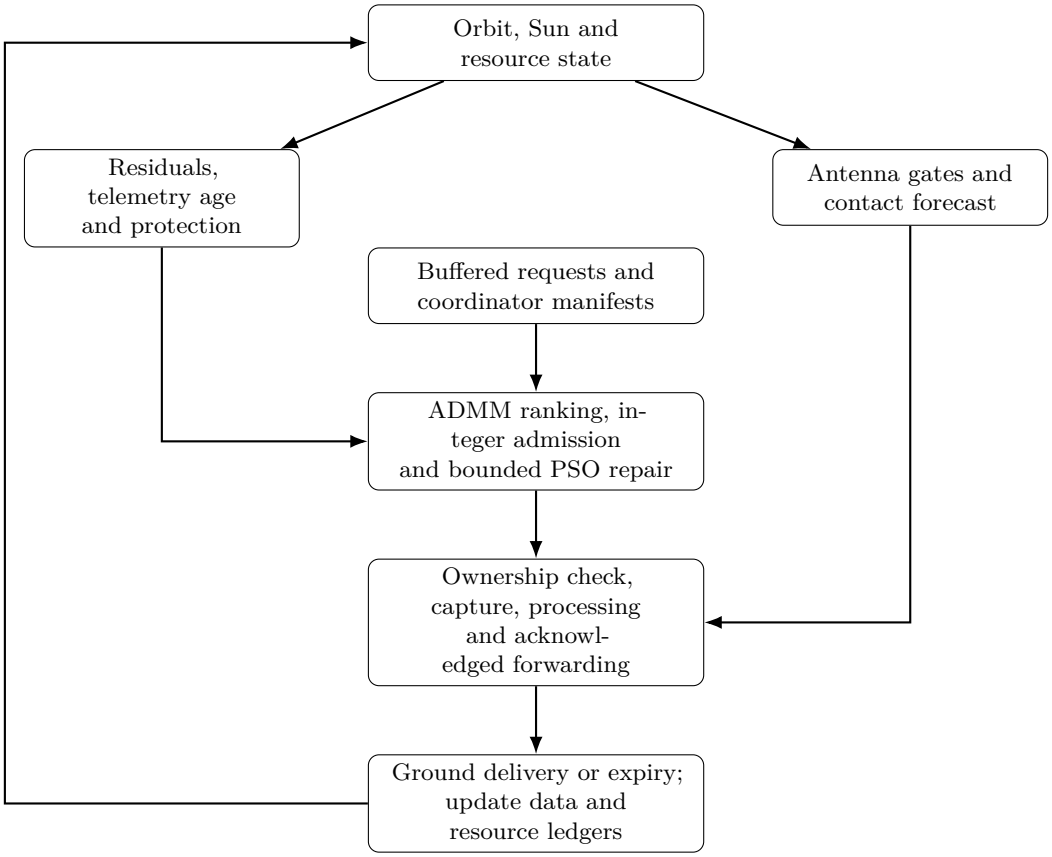


Figure 1: State, coordination and execution form a feedback loop. A fault changes available capabilities and triggers repair; completion requires a ground acknowledgement.

Each tick expires transient events, propagates the physical model, evaluates health, expires overdue tasks, advances control messages and updates workload. Planning normally runs every five seconds when ready work exists. Critical health and operator events can trigger repair in the current update. Capture, processing and transport then advance before ledgers are rebuilt. Changes in inter-satellite geometry are checked for reconstruction every 30 ticks.

4.2 Coordinator buffering

Only nodes capable of both computing and relaying are coordinator candidates. For $N > 1$, the implemented node score is

$$\omega_i = \frac{\sum_{l:\,(i,l)\in\mathcal{E}(t)} w_{il} + 30\big[(N-1) - \deg_i\big]}{N-1}, \qquad c^* = \arg\min_{i\in\mathcal{V}_{\text{eligible}}} (\omega_i, \mathrm{id}_i). \tag{4.1}$$

The missing-neighbor penalty prevents a disconnected node from appearing cheap merely because few edges remain. The existing coordinator remains stable until reconstruction or loss of eligibility; election is not repeated at every small score fluctuation. The lowest communication score selects the coordinator, not necessarily the task executor.

An ingress node retains a request until a weighted shortest path reaches the coordinator. A disconnected ingress buffer waits and retries. Losing a route therefore does not delete the mission request. The code models 2048-byte control messages and at most eight message-hop operations per radio per tick, or per second in the normal one-second loop. This control allowance is separate from the data-transfer budget.

4.3 Manifest sharing and ADMM ranking

The coordinator shares the task ID, requirements, priority, windows and deadline. Only nodes recorded as having received the manifest can be admitted for that task. Sharing need not finish at every node before reachable hosts can be considered. ADMM uses those eligible hosts and frozen resource rows.

Numerical ADMM exchanges are computed directly in process memory. Their reported payload estimate is

$$n_{\text{scalar}} = 3NMK_A, \tag{4.2}$$

for three $N \times M$ arrays per iteration. This is an accounting proxy, not a measurement of routed optimization traffic. The explicit request and manifest messages demonstrate control buffering, but do not constitute a distributed implementation of ADMM.

4.4 Ownership, relay handoff and task execution

Maintain a task record

$$\mathcal{L}_j = (\mathrm{id}_j, o_j, v_j, e_j, t_j^{\text{lease}}, \text{status}_j), \tag{4.3}$$

where v_j is the ownership version and e_j the coordinator epoch. A proposed assignment is accepted only if

$$\begin{aligned} e^{\text{req}} = e^{\text{current}}, \quad v^{\text{req}} = v_j, \quad o_j = \varnothing, \\ i \in \text{Shared}_j, \quad \text{Path}(c^*, i) \neq \varnothing, \quad \text{CanHost}(i, j) = 1. \end{aligned} \tag{4.4}$$

The owner is then set and the version incremented before the next candidate is considered. The lease field is set to the delivery deadline. This is an atomic compare-and-set style check within one authority, not a replicated quorum commit or a distributed clock-guarded lease protocol.

After acquisition, a CPU fault pauses processing on the current holder while preserving its progress. A radio fault may leave local processing available. A link outage reroutes the existing product. A storage fault makes the only copy inaccessible and fails the task. Unfinished acquisitions may be released and reassigned; already acquired products are not silently recaptured.

Each output has

$$n_j^{\text{pkt}} = \lceil D_j^{\text{out}} / 0.5 \rceil \tag{4.5}$$

logical packets. A packet is accepted only when its length and CRC32 match. Sequence numbers suppress repeated acknowledgement. The receiver commits the hop only after the full packet manifest is verified; the sender copy is then released. The simulated bytes are compact deterministic surrogates for the logical MB volumes. CRC checks detect modeled corruption; they provide no authentication against a malicious sender.

4.5 Ground routing and execution budgets

At runtime, for $D_{e,k}$ MB sent on edge e,

$$\sum_j D_{e,j,k} \le R_e \Delta t/8, \tag{4.6}$$

$$\sum_{e \text{ incident to } i} D_{e,k} \le 24\Delta t/8, \qquad \sum_{e \text{ incident to } g} D_{e,k} \le 48\Delta t/8. \tag{4.7}$$

Tasks share those budgets in priority order. These are aggregate byte-rate limits, not a detailed half-duplex MAC or interference model. Receiver memory is reserved before sending. Failed or partial packets are retried while a usable contact remains. If no route fits the deadline, valid data waits in its holder until a new route appears or the task expires.

5 Results and discussion

5.1 Evaluation scope and recorded run

The supplied release includes numerical, integration and browser checks. Its archived reports record 79 Python checks (31 integrated checks and 48 retained laboratory checks) and 16 browser workflow checks. These establish tested behavior, not statistical mission reliability. The following run statistics are transcribed from `evidence/full_orbit_final.json`, not newly inferred from screenshots.

Table 1: Selected defaults of the integrated simulator. All resource and search settings are demonstration parameters.

Parameter	Value
Satellites	Two groups of six; configurable up to 40 total
Initial altitude, inclination, spacing	550 km, 51.6°, 50 km adjacent chord
Battery capacity and protection reserve	100 Wh and 12 Wh
Thermal protection and admission limit	76°C and 74°C
Telemetry age limit	8 s
Storage and baseline control CPU	512 MB and 10% per node
Default capture and processing	5 s and 8×10^6 cycles
Default raw and output sizes	40 MB and 10 MB
ADMM budget and tolerance	20 iterations; $r, s \le 0.01$
PSO budget and repair scope	6 particles, 12 iterations, at most 8 tasks
Rare-fault attempt rate	$8 \times 10^{-5}\,\mathrm{s}^{-1}$ per configured satellite

Table 2: Archived 6000-s integrated run. Counts describe one trace and are not averages across independent experiments.

Quantity	Recorded value
Generated tasks	590
Delivered / expired / active	40 / 542 / 8
Acknowledged output at ground	400 MB
Route repairs	34
Rare stochastic satellite faults	4
Ticks with at least one eclipsed satellite	1934
Injected bit flips / partial packets	163 / 76
Packet retries	254
Graph reconstructions	469

The completion statistic must state its denominator:

$$\eta_{\text{generated}} = \frac{40}{590} = 6.78\%, \qquad \eta_{\text{terminal}} = \frac{40}{40 + 542} = 6.87\%. \tag{5.1}$$

The dashboard uses the second convention, excluding eight active tasks. The short default deadlines, repeated workload and single ground station leave many requests without a feasible delivery opportunity. The counts show that expiry is reported honestly. They do not isolate how much loss comes from contact geometry, admission choices, coordination delay or resource constraints; that requires tagged rejection reasons and controlled trials.

5.2 Fault detection and CUSUM

Hard protection checks $T_i \geq 76°\text{C}$ and $E_i \leq 12$ Wh immediately. Telemetry staleness is a distinct condition:

$$a_i^{\text{tel}} = t - t_i^{\text{tel}}, \qquad a_i^{\text{tel}} > 8 \text{ s} \implies \text{isolate node.} \tag{5.2}$$

To reveal unexpected heating, compare an actual thermal step with a prediction that excludes the injected heater:

$$r_{i,k}^{T} = T_{i,k}^{\text{actual}} - \widehat{T}_{i,k}^{\text{nominal}} + \nu_{i,k}, \qquad \nu_{i,k} \sim \mathcal{N}(0, 0.02^2), \tag{5.3}$$

$$S_{i,k} = \max\left(0, S_{i,k-1} + \frac{r_{i,k}^{T}}{0.15} - 0.6\right), \qquad S_{i,k} \geq 12 \implies \text{latch alarm.} \tag{5.4}$$

The divisor is an illustrative normalization, not the actual noise standard deviation in this version. CUSUM accumulates positive departures so a persistent shift can trigger before an absolute temperature limit [9]. It is one-sided: it does not detect every cooling anomaly or diagnose the failed component. Its thresholds require calibration over healthy eclipse, workload and attitude transitions.

Full isolation follows severe thermal, reserve, storage, offline or stale-data conditions. A CPU fault removes computation but can preserve relaying; a radio fault removes relaying but can preserve computation. An attitude fault removes observation capability. Recovery from full isolation requires cleared causes, three fresh safe samples and temperature below 70°C. Optional demonstration recovery can reset resources; this reset must be distinguished from natural thermal cooling or battery recharge.

Rare fault attempts use exponential interarrival times with total rate $\Lambda = N\lambda_f$. Thus

$$\Pr\{\text{at least one attempt in } \Delta t\} = 1 - e^{-N\lambda_f \Delta t}. \tag{5.5}$$

The code then chooses a currently fault-free node, if one exists. This models demonstration fault injection rather than a calibrated independent component reliability law. The rare-fault switch is separate from packet disturbances.

5.3 Allocation and optimization

ADMM plots should show both residuals, termination status and host fractions. The ownership and resource table should show which fractional preferences became actual reservations. For a controlled fault case, compare the greedy incumbent and best PSO decoder scores:

$$\Delta F = F_{\text{best}} - F_{\text{incumbent}} \geq 0 \tag{5.6}$$

on the same frozen candidate set. This search property does not guarantee that the finally executed mission improves after state changes or later revalidation. The archived long run contains no task



Figure 2: Fault evidence and capability isolation. Report detection delay from the injection timestamp, separately from the time to a repaired plan and delivery.

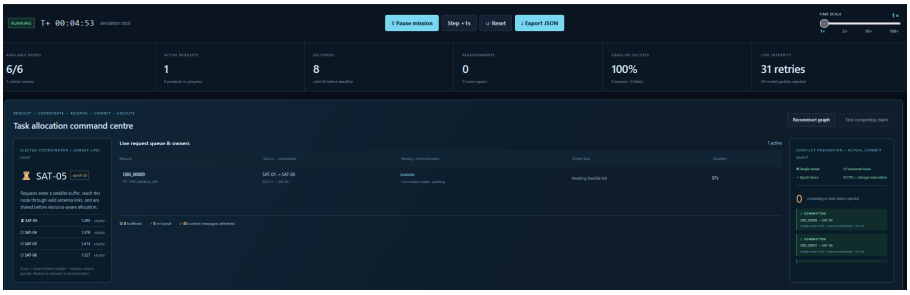


Figure 3: From fractional ranking to executable task admission. Mark the injected fault and show the released acquisition and replacement owner, if applicable.

reassignments, so reassignment behavior must be demonstrated using a targeted acquisition-fault scenario rather than claimed from that run.

Useful comparison cases are resource-aware greedy admission, ADMM plus integer admission, and the same method with bounded PSO. Use identical arrivals, fault traces and link conditions. Report priority delivered, deadline success, CPU time, energy and coordination overhead across several seeds. Small exhaustive or integer-programming instances can quantify the heuristic gap. The present evidence does not establish that gap.

5.4 Ownership conflict simulation and coordination

A conflict test submits a competing or stale claim after a task has an owner. The expected result is rejection with unchanged owner, version and resource reservations. An epoch test changes coordinator and verifies that a successor without the manifest first receives a retained copy before it assigns work. An ingress-partition test verifies that a disconnected request remains buffered.

One stored owner per task follows from the current serial authority. A distributed deployment needs persistent replication, membership and recovery rules. Numerical consensus does not supply those rules. A lease also cannot prove that a silent previous owner never took an image; strict one-attempt semantics would require durable attempt tokens at the actuator interface.

5.5 Network routing

Graph-history views preserve the seeded initial graph and later reconstructions. Available edges are gated by health and geometry; their weights retain the calibration seed and update the physical/resource terms. A dark backbone is a Kruskal minimum spanning forest:

$$\mathcal{F}^* = \underset{\mathcal{F} \text{ spans each available component}}{\arg\min} \sum_{e \in \mathcal{F}} w_e. \tag{5.7}$$

It is a display summary of available connectivity, not a PSO output or a restriction forcing traffic onto tree edges. Highlight the actual task route with another style so backbone, availability and execution remain distinct.

The archived 34 reroutes and 400 MB of acknowledged output demonstrate that alternate delivery paths were exercised in that trace. Corruption and partial-packet counters show that the transport fault mechanisms were active; they are not bit-error-rate estimates. A reconstruction can occur within one simulation update, but this is not zero physical recovery latency. Measure wall-clock planning time, simulated delay to the next contact and end-to-end delivery delay separately.

6 Applications

6.1 Space observation and onboard data delivery

For fire, flood or storm monitoring, a task specifies an observation instance, quality requirement and delivery deadline. Repeated instances provide updates as the event changes. The allocator can reject an apparently well-powered observer when no delivery path exists, and a relay can carry an existing product without repeating the observation. Optical and GNSS-related requests can share an event ID

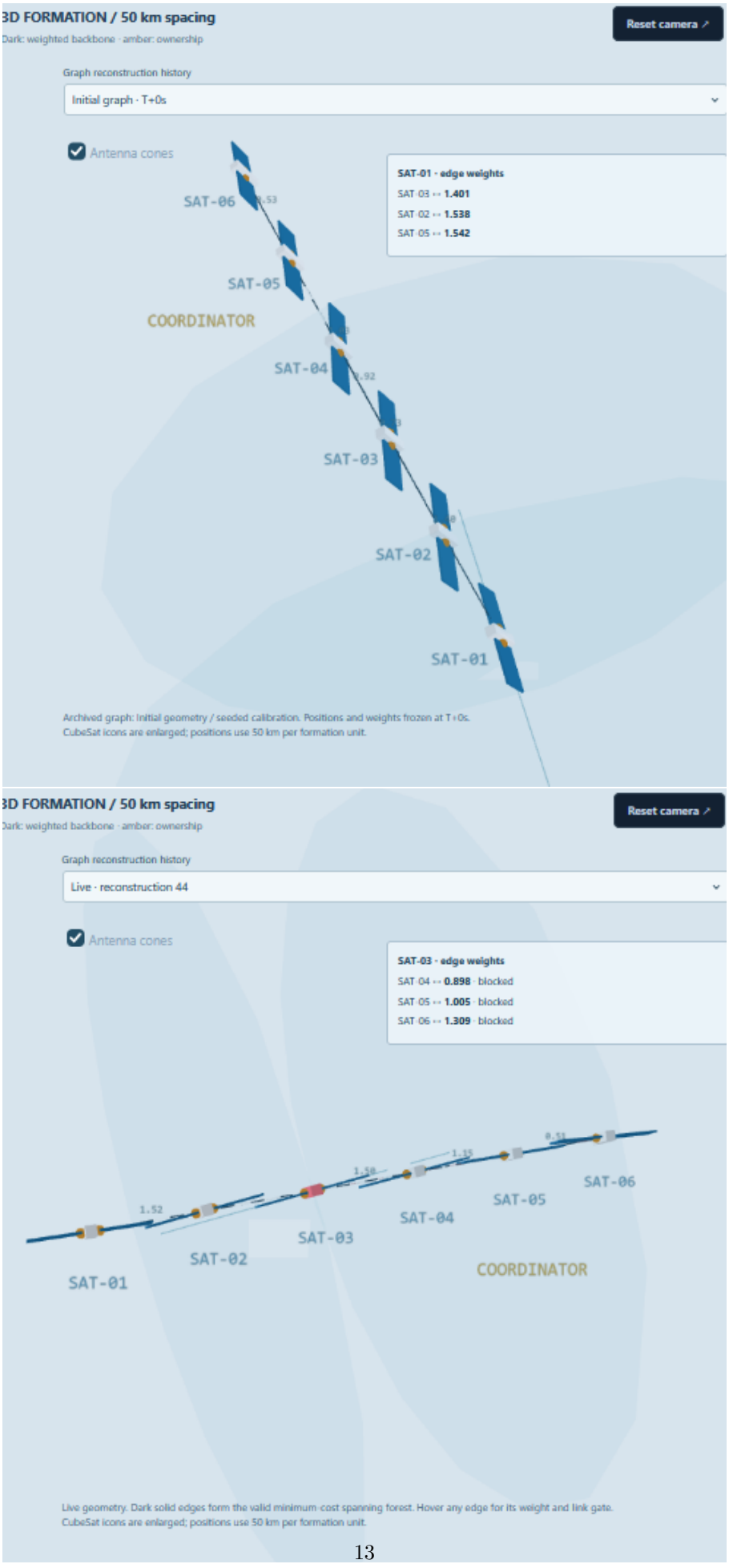


Figure 4: Graph reconstruction preserves the original seeded view and records later epochs. A partition produces a forest; no unavailable physical edge is invented.

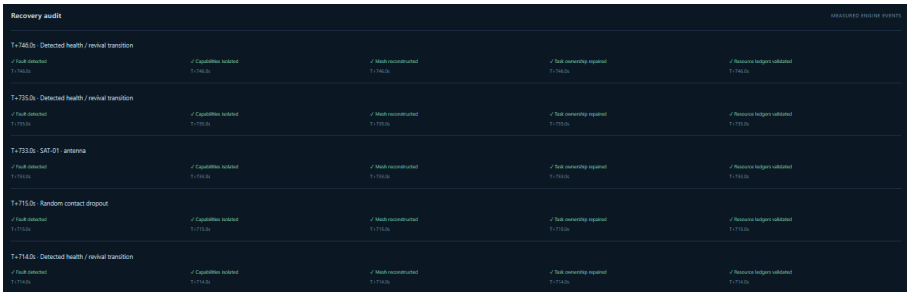


Figure 5: Causal forwarding through available contacts. A link failure retains the sender copy and repairs delivery without repeating the acquisition.

while retaining distinct task IDs and resource profiles; the simulator represents their workflow, not their sensor retrieval physics.

The same structure is relevant to hosted payloads and spacecraft clusters with intermittent ground access. A payload fault changes observation eligibility, a CPU fault changes processing, and a radio fault changes contact availability. Eclipse-aware charging makes the energy cost of processing and forwarding depend on when the work occurs. Extension to multi-orbit constellations requires genuine ephemerides, camera geometry, contact plans and onboard timing measurements.

6.2 Industrial edge sensing and inspection

In a plant or remote infrastructure network, a task may mean acquiring an inspection image, evaluating a vibration segment, classifying a thermal alert or delivering an equipment report. Sensor nodes and gateways have limited CPU, storage, power and communication. The task should be considered complete when its required result reaches the plant sink, not when a measurement is merely sampled.

The coordination interface can be reused with a different physical model:

$$\mathcal{F}_i = \begin{cases} \mathcal{F}_i^{\text{space}}, & \text{orbit, eclipse and antenna constraints,} \\ \mathcal{F}_i^{\text{industrial}}, & \text{plant access, thermal and network constraints.} \end{cases} \tag{6.1}$$

Scheduled wireless access or mobile-gateway visits replace orbital contacts. Plant heat-transfer and battery models replace the spacecraft equations; convection may then be relevant. Local anomaly evidence can support investigation, as discussed for industrial edge agents by Singh et al. [4], but a thermal CUSUM alarm is not evidence of cyber intrusion. This industrial use is an adaptation of the framework, not a validated industrial deployment.

7 Conclusion

The central modeling choice is to treat acquisition, processing, retained data and delivery as one obligation. This explains why a low link cost alone cannot select an executor, why a valid image should survive a temporary route failure, and why expiry must remain visible. The simulator combines geometric access, resource screening, ADMM ranking, bounded PSO repair and acknowledged forwarding while separating work ownership from data possession.

The recorded run demonstrates exercised fault and transport mechanisms but also substantial deadline loss. The next experimental question is which combination of contacts, workload and repair policy improves delivered value at acceptable compute and communication cost. Answering it requires controlled baselines and calibrated models, followed by a truly distributed implementation of the coordination and ownership layers.

8 Caveats

1. **Decentralized workflow is emulated.** Agent state, ADMM arrays, coordinator election and task authority share one process. Message buffers demonstrate selected communication behavior; they do not prove asynchronous convergence or fault-tolerant replicated ownership.

2. **Orbital and solar fidelity is limited.** The model has Earth J_2, a spherical occulting Earth, a circular annual Sun direction and a cylindrical shadow. It omits drag, third-body gravity, penumbra, solar distance variation and maneuver dynamics. The epoch is illustrative.

3. **Observation access is synthetic.** Windows are not derived from target coordinates, terrain, camera field of view or clouds. Image quality is fixed. A connected mesh does not imply target visibility.

4. **Power and thermal coefficients need calibration.** Solar-wing floors, fast thermal response and per-MB energy charges are demonstration choices. Earth infrared, albedo, detailed conversion losses, battery aging, receiver energy and radio/payload heat are not fully resolved. The no-solar admission proxy is not a certified eclipse-survival calculation.

5. **Admission is not complete future scheduling.** CPU and storage reservations are explicit, but admitted jobs do not reserve a complete shared future contact schedule or an exact aggregate battery trajectory. Runtime contention and forecast mismatch can invalidate predicted delivery. A full robust scheduler needs joint time-indexed reservations and uncertainty bounds.

6. **Antenna and network models are coarse.** Box occlusion and cosine gain replace measured patterns. Rates are surrogate values; Doppler is primarily a diagnostic/cost term. No detailed RF link budget, interference or MAC implementation is supplied. Control traffic has a separate allowance and does not debit the data channel's byte ledger.

7. **Finite search is heuristic.** ADMM solves a frozen convex relaxation to a tolerance or iteration cap. Integer decoding and 78 PSO evaluations do not establish a global optimum. Simple shortest path is not NP-hard, and three route candidates can be compared directly.

8. **Recovery does not reconstruct lost data.** CPU outages pause local processing; products wait through temporary radio outages. No durable replication or processing migration is implemented. A storage fault can destroy access to the only copy. Optional resource resets represent operator recovery in the demonstration.

9. **Ownership has no distributed exactly-once proof.** Epoch and version guards prevent conflicting serial claims in this simulator. A real partitioned system needs replicated authorization and an explicit uncertain-attempt policy. CRC32 and hashes do not replace authentication.

10. **Validation evidence has a narrow scope.** Archived test passes and one long trace support the recorded software behavior. They do not establish flight reliability, superiority to baselines, calibrated detector false-alarm rates or industrial cybersecurity performance. Multi-seed comparisons and hardware measurements remain necessary.

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